



How to read a paper in AI in medicine

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What are different types of papers in AI in medicine?

- **Algorithm development**
- **Applications**
- **Systematic reviews**
- **AI in Ethics, implications of AI in medicine**

Deep Residual Time-Series Forecasting: Application to Blood Glucose Prediction

Education and Career Development

Generating Rare Surgical Events Using CycleGAN: Addressing Lack of Data for Artificial Intelligence Event Recognition



Artificial Intelligence in Medicine

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Reinforcement learning application in diabetes blood glucose control: A systematic review

Ethics of artificial intelligence in medicine

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PMCID: PMC7615805 EMSID: [EMS194524](#) PMID: [38527299](#)

What are different approaches to reading a paper?

- **One approach: Reading from start to finish, sequentially**
- **Skimming: Quickly glancing through the paper, to get a rough idea**
- **The Three-Pass Approach**
 - helps with prioritizing the papers and its different parts
 - focuses on relevant details

First Pass

- **5 – 10 minutes**
- **Get a bird's-eye view of the paper.**
- **Steps:**
 - **Read title, abstract, introduction, and conclusion.**
 - **Skim headings and references.**
 - **Identify the 5 Cs: Category, Context, Correctness, Contributions, Clarity.**

The 5 Cs

- **Category:**
 - What type of paper is this? A measurement paper? An analysis of an existing system? A description of a research prototype?
- **Context:**
 - Which other papers is it related to?
 - Which theoretical bases were used to analyze the problem?
- **Correctness:**
 - Do the assumptions appear to be valid?
- **Contributions:**
 - What are the paper's main contributions?
- **Clarity:**
 - Is the paper well written?

Second Pass

- Up to 1 hour
- Read the paper more carefully, excluding proofs
- Pay attention to:
 - Figures, graphs, and results (check for clarity and statistical significance).
 - References (identify important related works).
- Aim: Understand the main argument and evidence.

What if you still don't understand the paper?

Reasons:

- You are new to the field and there are a lot of unfamiliar terminology
- Complex proofs or experimental techniques
- Poor writing, or too many references
- Simply being too tired at the end of the day!

What if you still don't understand the paper?

Choices:

- **Set it aside – If it's not essential to your research.**
- **Revisit later – After gaining more background knowledge.**
- **Push through – Attempt a third pass for a deeper dive.**

Third Pass

- **Used for thorough comprehension**
- **Requires re-implementing the paper's work mentally.**
- **Can take several hours, but is essential for mastery.**

- **Steps:**
 - **Identify assumptions, limitations, and missing citations.**
 - **Compare the author's approach with alternative methods.**
 - **Critically evaluate the study's strengths and weaknesses.**

This process also relies heavily on your personal preferences, and research habits

What questions should you ask when reading a paper?

It's about **critical thinking**
and asking a lot of **whys** and
hows

Article | [Open access](#) | Published: 04 April 2025

Deep learning prediction of mammographic breast density using screening data

[Chen Chen](#), [Enyu Wang](#), [Vicky Yang Wang](#), [Xiayi Chen](#), [Bojian Feng](#), [Ruxuan Yan](#), [Lingying Zhu](#)  & [Dong Xu](#) 

[Scientific Reports](#) **15**, Article number: 11602 (2025) | [Cite this article](#)

3654 Accesses | **3** Citations | **1** Altmetric | [Metrics](#)

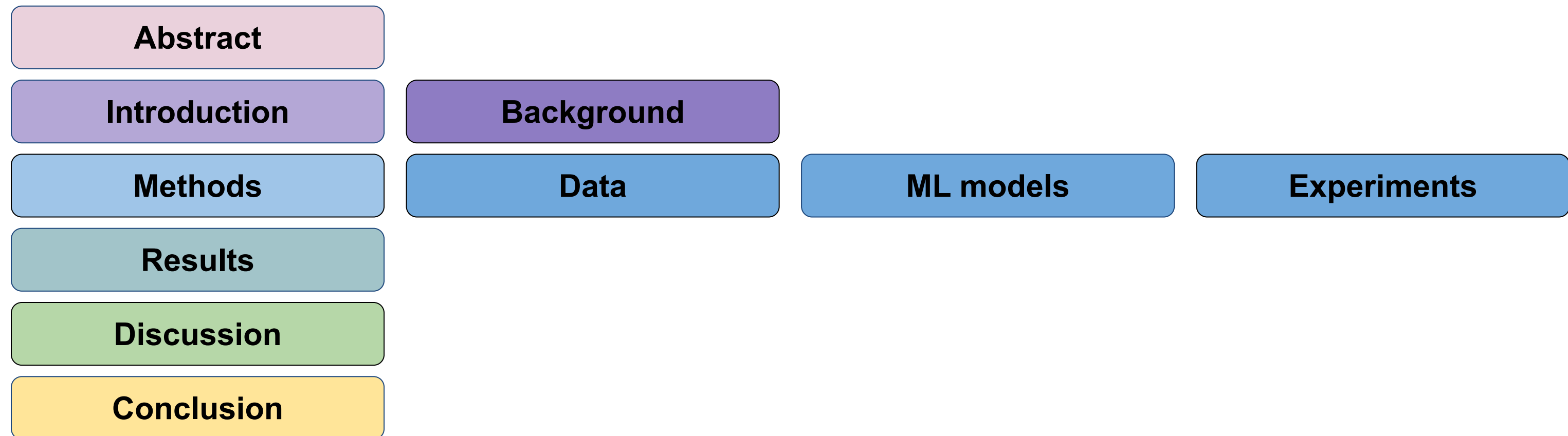
Chen, C., Wang, E., Wang, V. Y., Chen, X., Feng, B., Yan, R., Zhu, L., & Xu, D. (2025). Deep learning prediction of mammographic breast density using screening data. *Scientific Reports*, 15(1).

<https://doi.org/10.1038/s41598-025-95275-5>

Details of the paper

Date:	4 April 2025
Journal:	Nature Scientific Reports
Impact Factor	4.3 (5-year impact factor)
Citations:	5 citations
Location/Research Group:	Taizhou Key Laboratory of Minimally Invasive Interventional Therapy & Artificial Intelligence
Who are the authors?:	MD/PhDs
1st author(s)	Chen Chen
Corresponding author(s)	Lingying Zhu and Dong Xu

An AI Application paper structure in general



Title of the paper

- **Is it meaningful? Does it state what the problem and solution are?**
- **What is the purpose of the paper?**
 - Prediction?
 - Diagnosis?
 - Classification/Triage?
 - Treatment recommendation?
- **Is this a problem worth solving?**

Details of the paper

Topic:

Purpose:

Prediction?

Diagnosis?

Classification/Triage?

Treatment recommendation?

Why? What is the clinical value?

Where does it fit in the clinical workflow?

prediction of mammographic breast density using screening data

Prediction

- Classify **mammographic breast density into one of four BI-RADS categories**
 - **almost entirely fatty**
 - **scattered fibroglandular**
 - **heterogeneously dense**
 - **extremely dense**

Multi-class classification (4-class)

Breast density is an independent **breast cancer risk factor**

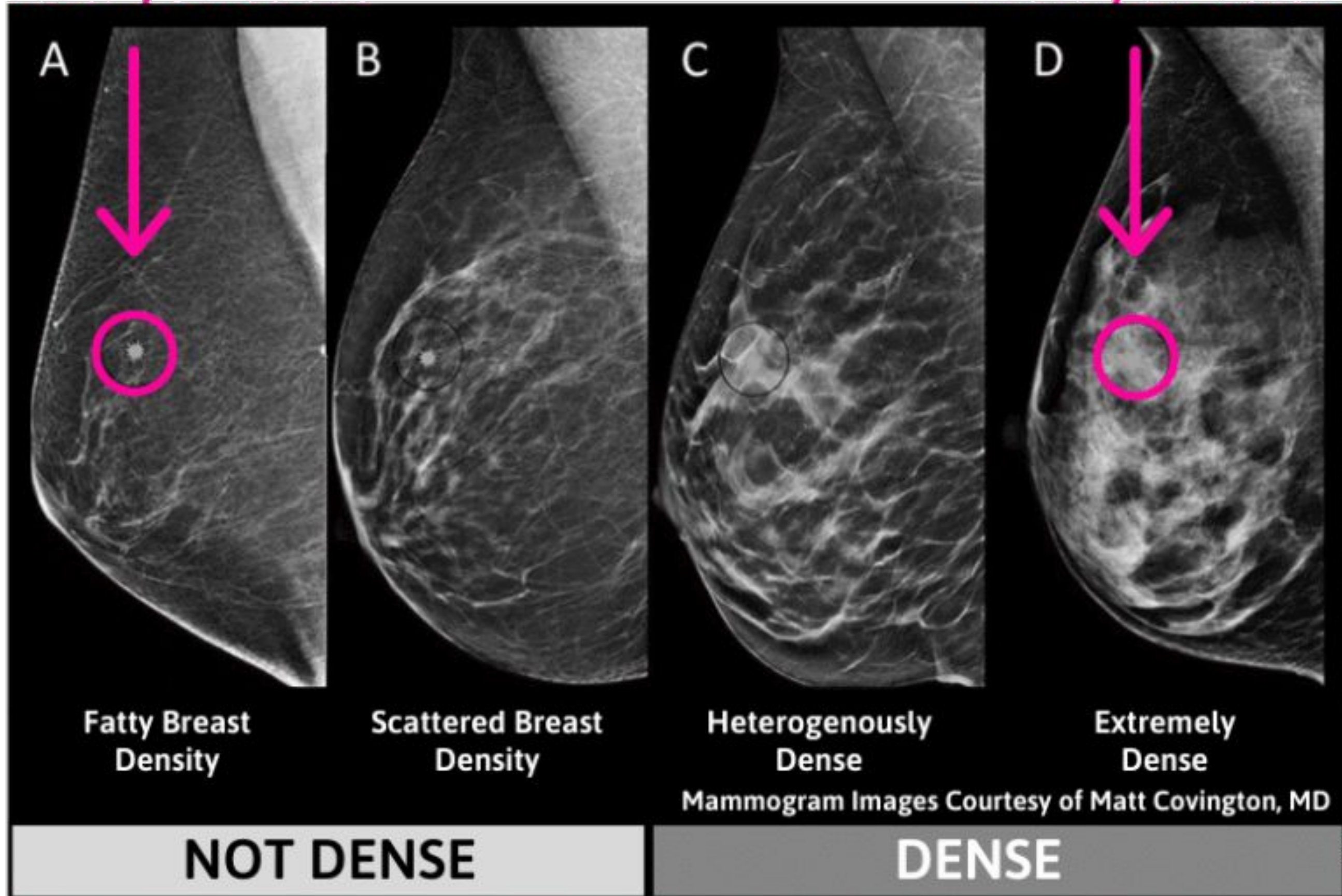
This model provides faster, more consistent, objective classification to standardize density reporting and flag high-risk women

How does classifying density in mammograms help with cancer detection?

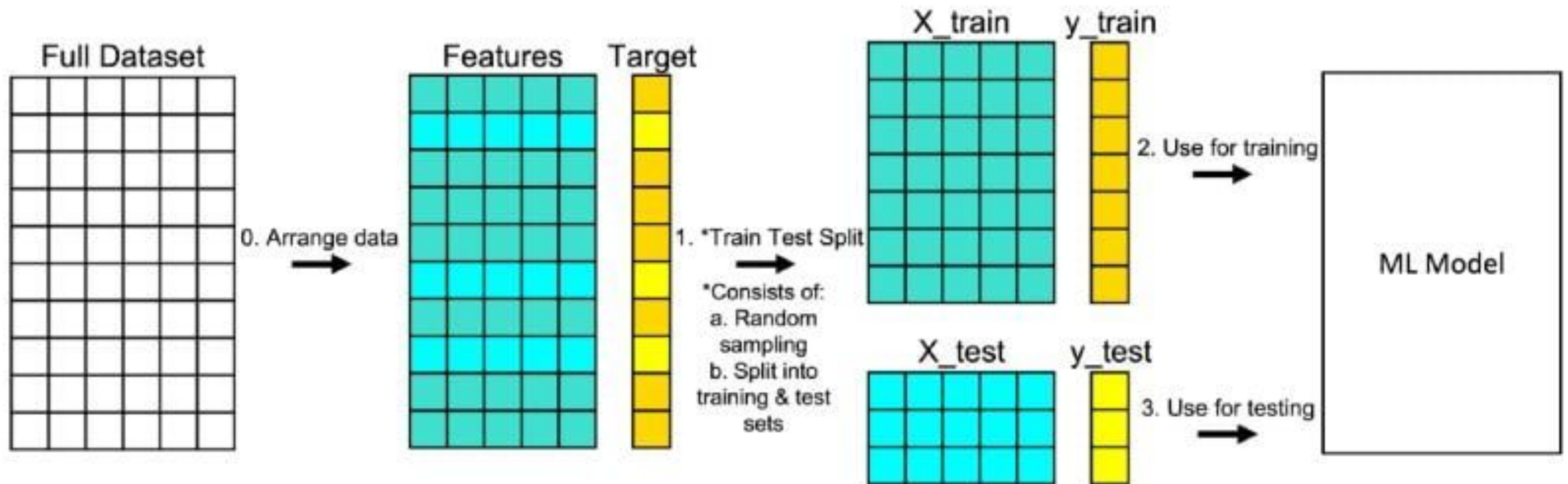
- **Normal dense breast tissue looks white on a mammogram, and so do tumours. So dense tissue can hide tumours.**
- **Women with dense breasts have a higher inherent risk of developing breast cancer**
- **Accurate density classification guides supplemental screening decisions (e.g., whether to add ultrasound or MRI for dense-breast patients)**
- **Supports earlier detection and enables faster triage at scale**

Now you see it

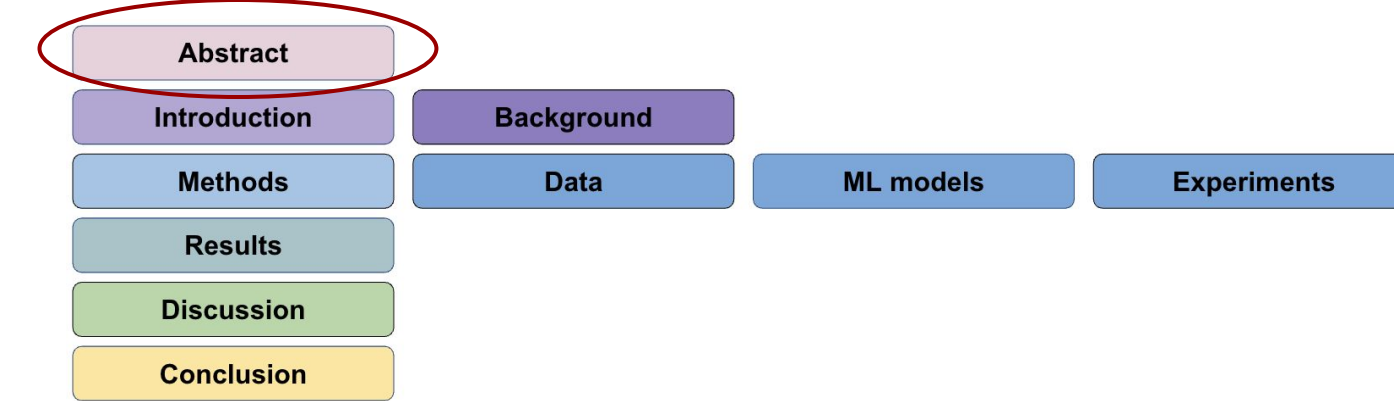
Now you don't



Reminder: Machine learning pipeline



Abstract



- **Are the problem, method, and solutions clearly stated?**

Problem

- **subjective and variable radiologist assessment of breast density (BI-RADS categories)**
leads to inconsistent cancer risk evaluation

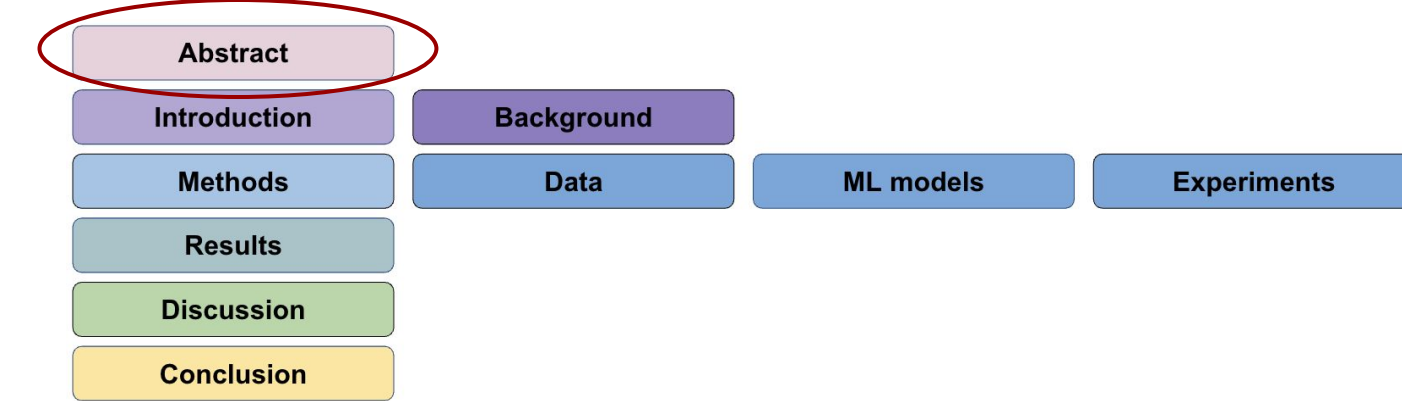
Method

- **four CNN-based deep learning models (ResNet50, InceptionV3, DenseNet121, Xception)**
- **trained on 57,282 mammograms from 9,621 women,**
- **evaluated using Average Precision (AP)**

Solution

- **InceptionV3 outperformed radiologists in accuracy and consistency, with strong AP values across all four density categories, validated on an external test set**

Details of the paper



Medical AI developed without input from a clinician is at risk of being clinically impractical.

You can develop an AI model to do a task very well, but can it actually change outcomes?

Target

Women undergoing mammographic breast cancer screening

Goal

Reduce inter-radiologist variability in breast density scoring

Improve speed and consistency of density assessment at scale

Outcomes

Reduced subjectivity

Reduced costs

Introduction

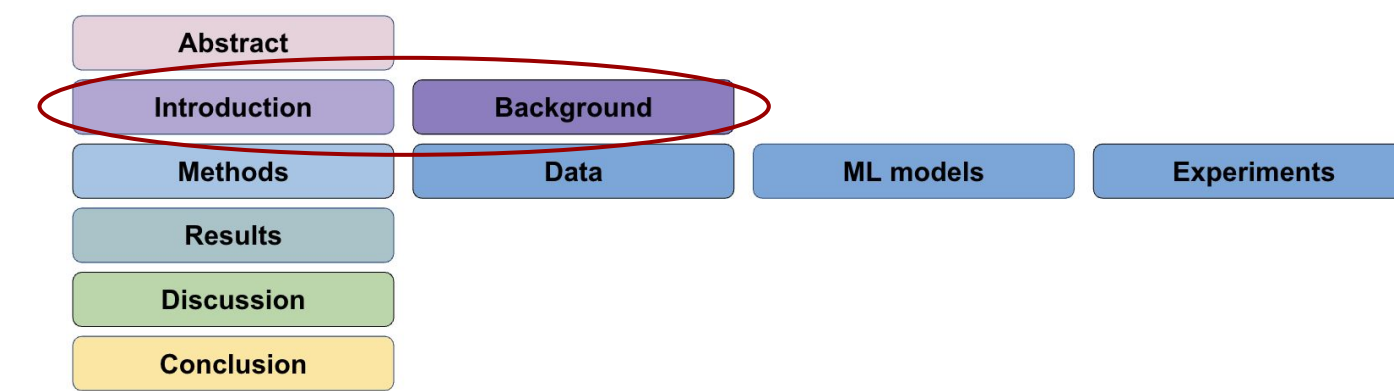


- What is the main problem that is addressed?
- Are there any over claims?

Main addressed problem

- BI-RADS for cancer assessment is highly subjective – radiologist agreement on dense breast percentage ranges from 6.3% to 84.5%
- This variability leads to inconsistent cancer risk stratification and potentially missed cancers in dense breasts
- Existing alternatives (MRI, ultrasound) have limitations:
 - MRI is costly and immobile;
 - ultrasound has poor image quality in high-density breasts
- There is a clinical need for objective, automated, scalable density assessment

Background



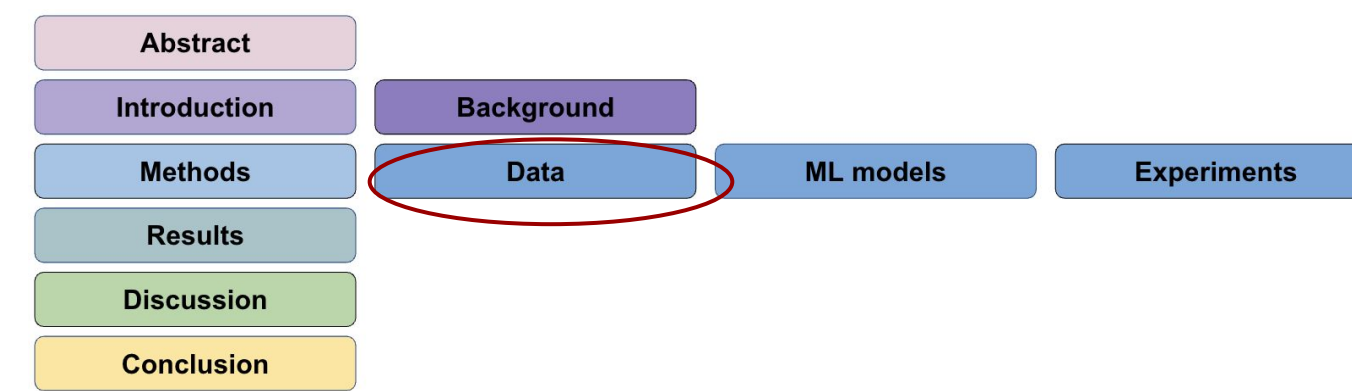
- Might be referred to as “related work” in some papers, or be part of the introduction
 - What is the knowledge gap?
 - How does the work build on top of the prior work?
-
- knowledge gap
 - previous work used relatively small datasets
 - previous work relied on non-consensus, single-radiologist labels as ground truth, introducing subjective bias into training
 - previous work did not rigorously compare DL performance against radiologists of varying experience levels

Dataset



- What are the features, sample size, preprocessing methods, etc?
- Might there be data leakage?
- How is the data split? is there data kept apart for validation of the experiments?
- How diverse is the dataset? Are there underrepresented groups?
- What are the potential underlying biases?
- Is this the right (type of) data?

Details of datasets



Data is the most important!

- What data is being collected?
- How much? Are there enough data points?
- From a single population or multiple?
- From a single institution or multiple?

For generalizability, data ideally should be from multiple populations and institutions

- Full-field digital mammographic images (MLO and CC views for each breast)
- Each image labeled with one of four BI-RADS breast density categories (a-d) by consensus of senior radiologists
- Basic clinical metadata: age, surgical history
- 57,282 mammograms from 9,621 women
- Split 7:2:1
- Single institution: Taizhou Cancer Hospital, China
- External validation set: only 500 cases from Kaggle/RSNA
- Skewed toward younger, premenopausal women

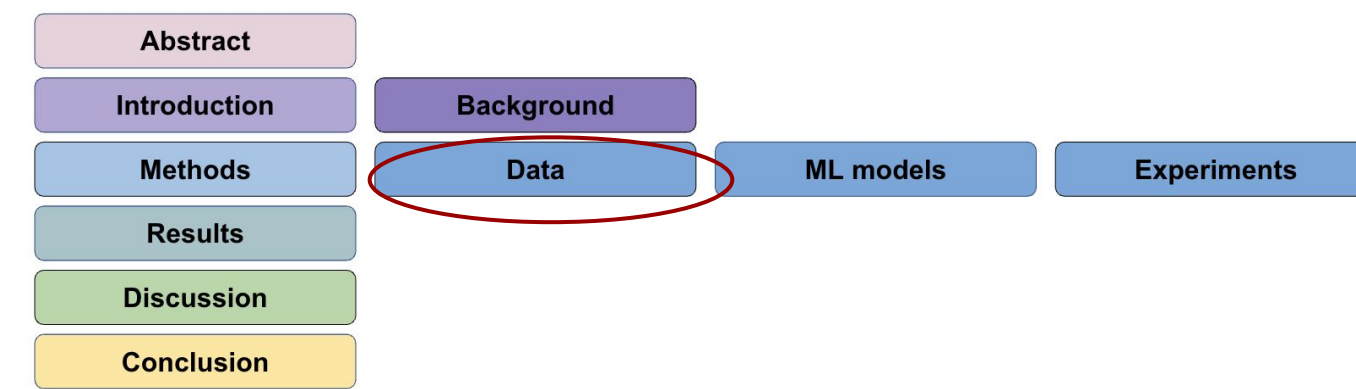
Details of datasets

Garbage in garbage out

Accurate labeling?

Missing data points?

Is there class imbalance?



- Labeling is relatively robust: consensus assessments from a panel of senior radiologists (expert committee) used as gold standard
- Divergent assessments resolved by a third senior radiologist, reducing individual bias
- Missing data is acknowledged but minimal:
 - 9 cases in training (0.13%), 1 in validation (0.05%), none in test set
- the meta data is limited
- significant imbalance: Category a (fatty): only 0.6% of cases – severely underrepresented

Details of datasets

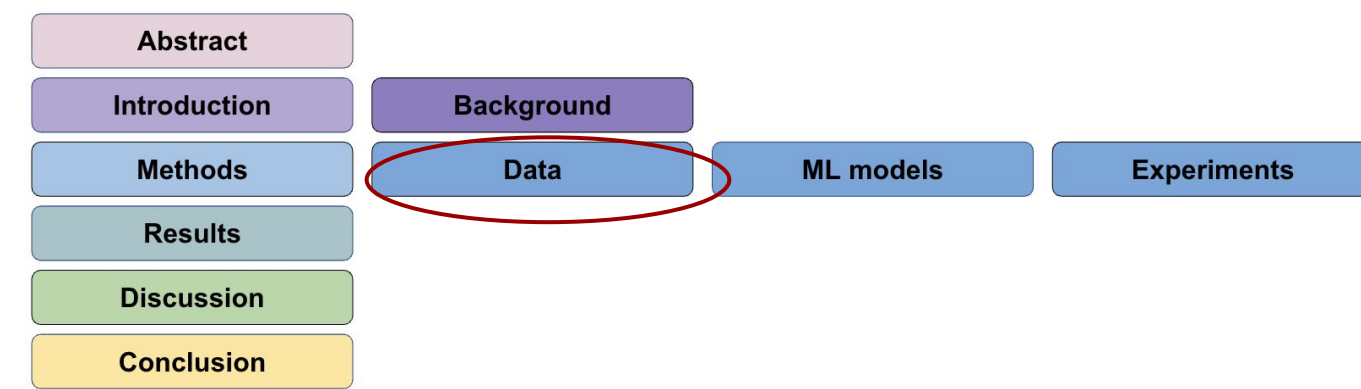
- Training
- Validation
- Test

The terms “Validation” and “Test” are often used incorrectly.

Validation sets are used for testing **during training**

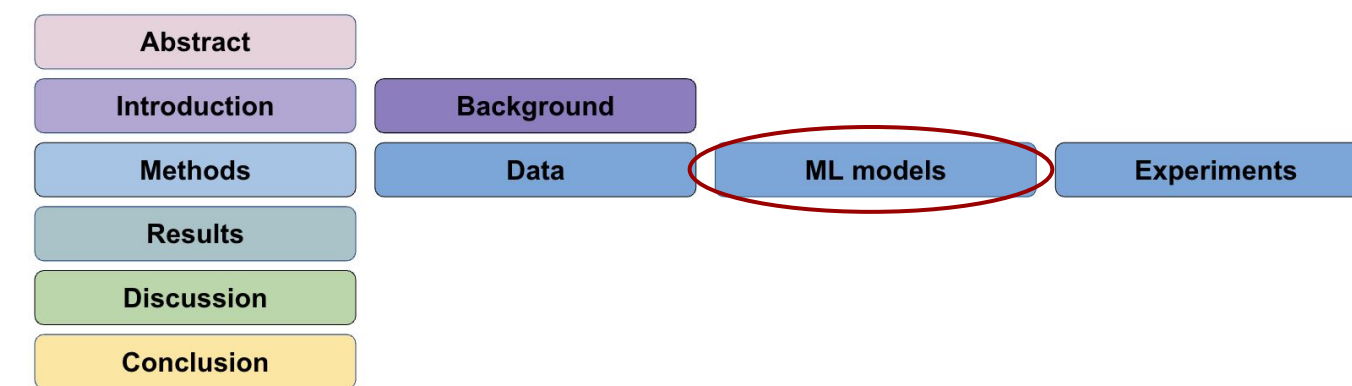
Test sets are used **to test the model after training is complete**

- Training (70%)
- Validation (20%)
- Test (10%)



Characteristic	Training Set	Validation Set	Test Set
No. of images	38,551	12,316	6415
No. of patients	6734	1924	963
Age(y)	46.60 ± 7.73	46.32 ± 7.74	46.63 ± 7.87
missing value	9(0.13)	1(0.05)	0(0)
<40	1040(15.44)	350(18.19)	174(18.07)
40–49	3600(53.46)	980(50.94)	491(50.99)
50–59	1699(25.23)	485(25.21)	229(23.78)
60–69	325(4.83)	96(4.99)	60(6.23)
≥70	61(0.91)	12(0.62)	9(0.93)
Senior radiologists-assessed breast density			
Category a (almost entirely fatty)	43(0.64)	14(0.73)	5(0.52)
Category b (scattered areas of fibroglandular tissue)	752(11.17)	148(7.69)	90(9.35)
Category c (heterogeneously dense)	3302(49.03)	758(39.4)	377(39.15)
Category d (extremely dense)	2637(39.16)	1004(52.18)	491(50.99)

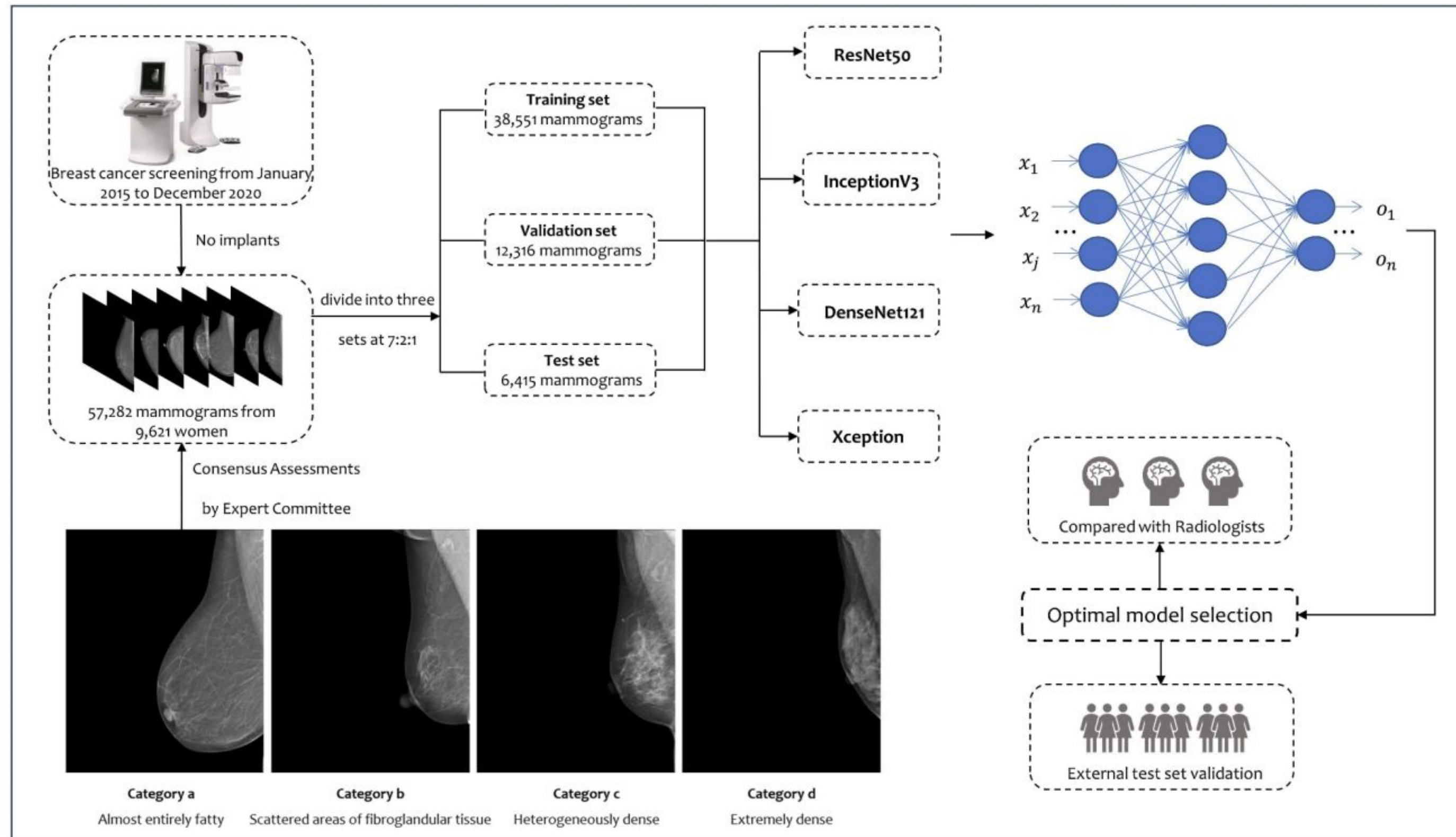
Model



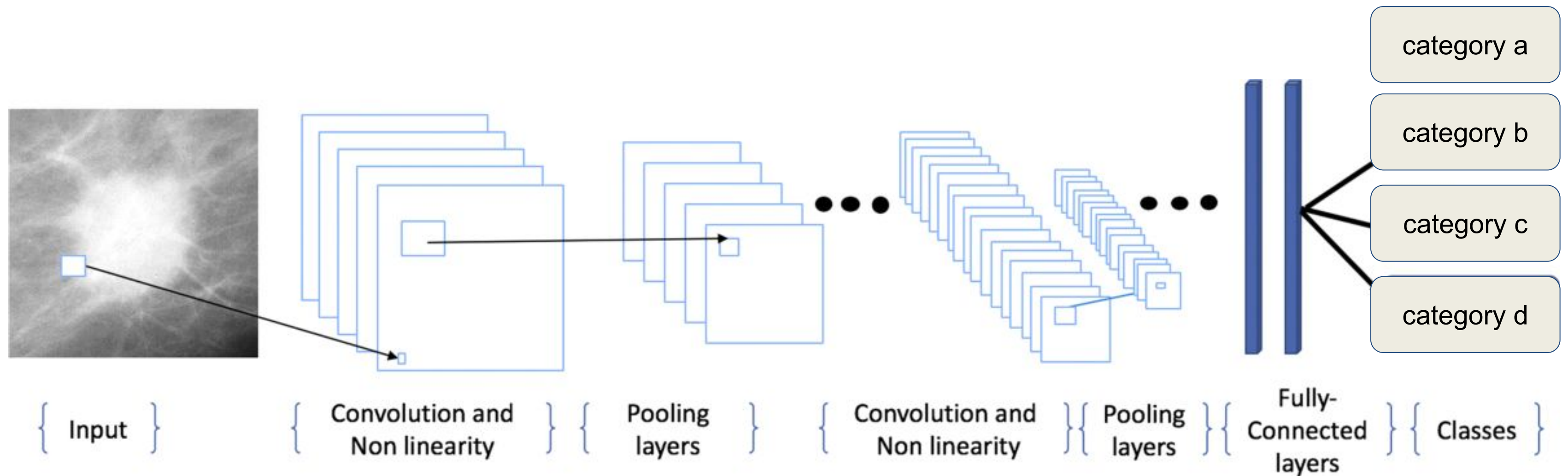
- **What is the model's architecture?**
- **Are there any algorithmic developments, and if so, how are they useful?**
- **Training process and hyperparameters?**
- **How is randomness handled?**
- **Is the model compared with other models? If so, how?**
- **Has this model been used before? If yes, in what context? Is this transferrable?**

CNNs for Medical Images

Abstract		
Introduction	Background	
Methods	Data	ML models
Results		Experiments
Discussion		
Conclusion		



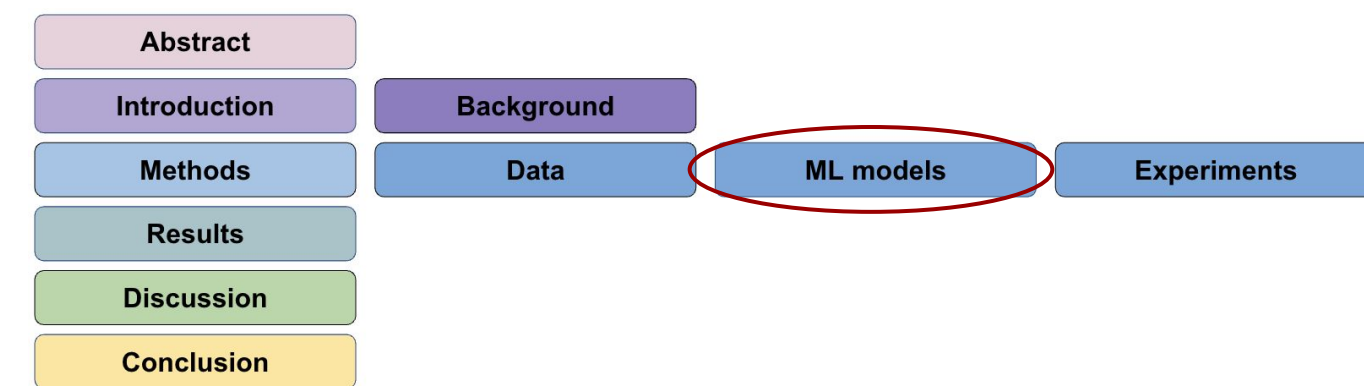
Recap: Convolutional Neural Networks



Details of the models

Are they appropriate for the dataset and purpose?

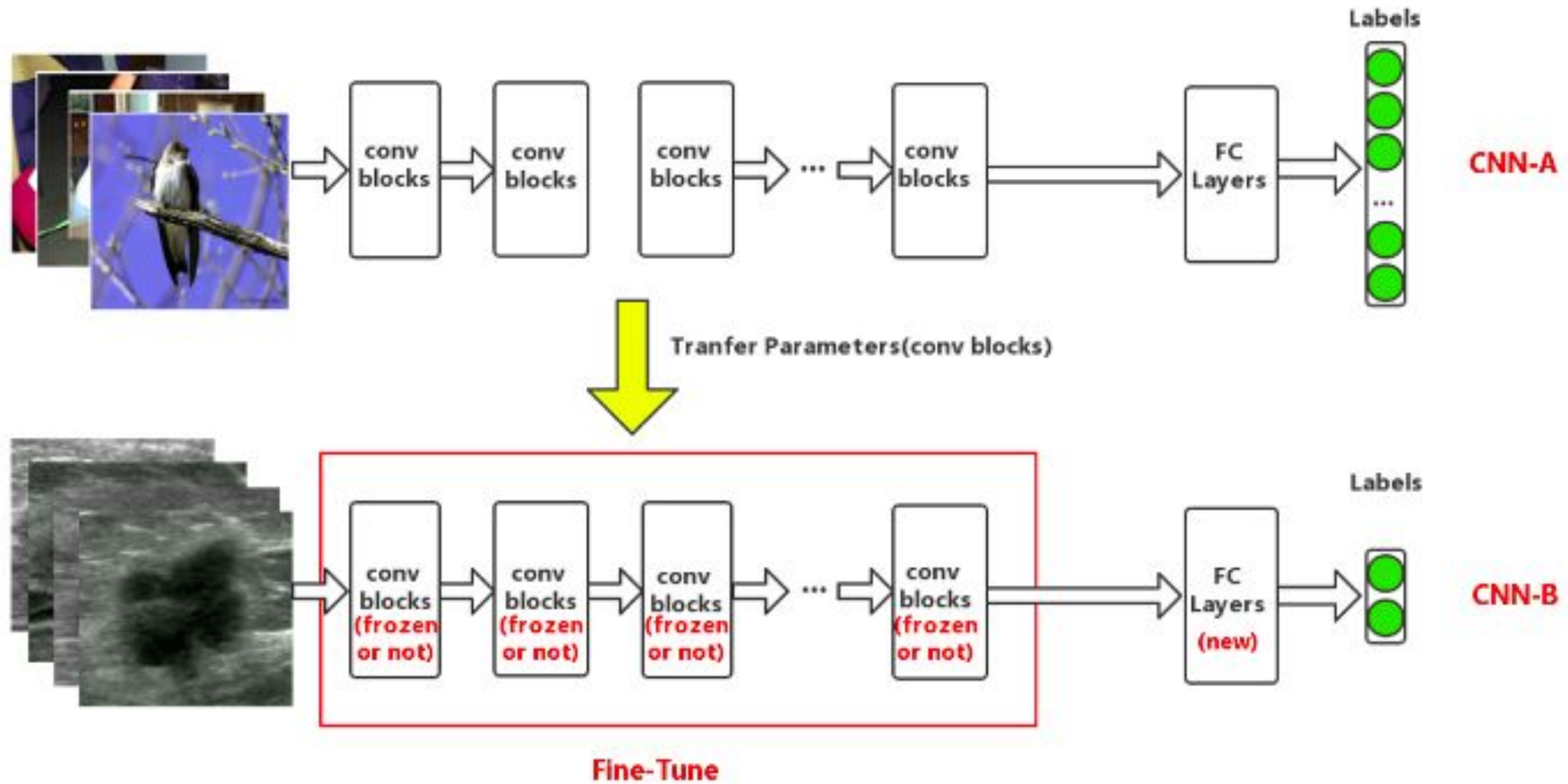
- CNN?
- Deep Learning?
- Reinforcement learning?
- Simple regression?
- Decision trees?



Four CNN-based architectures, all pre-trained on **ImageNet** (transfer learning):

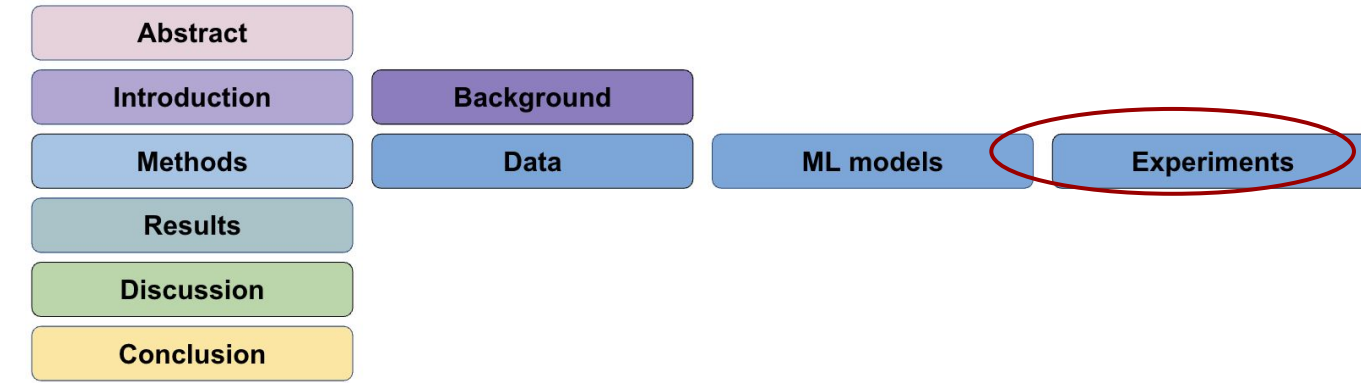
- **ResNet50**
 - residual connections enabling very deep networks
- **InceptionV3**
 - inception modules capturing multi-scale spatial features
- **DenseNet121**
 - densely connected layers promoting feature reuse
- **Xception**
 - depthwise separable convolutions for computational efficiency

What is transfer learning?

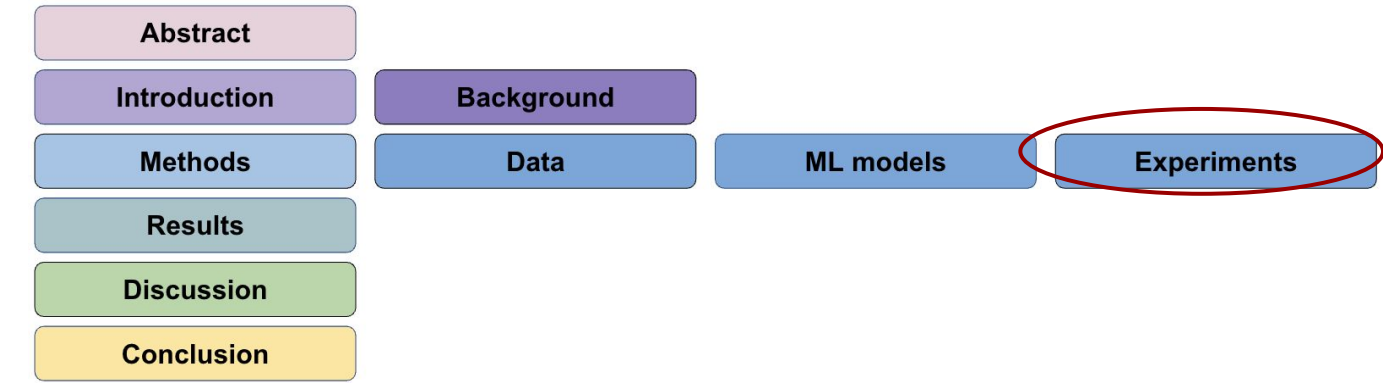


Experiments

- What is the rationale?
- What are the hypotheses?
- What are the assumptions?



Evaluation methods



- Are they clinically relevant?
- Are they representative of the AI model's performance?

Details of evaluation methods

What metric was chosen?

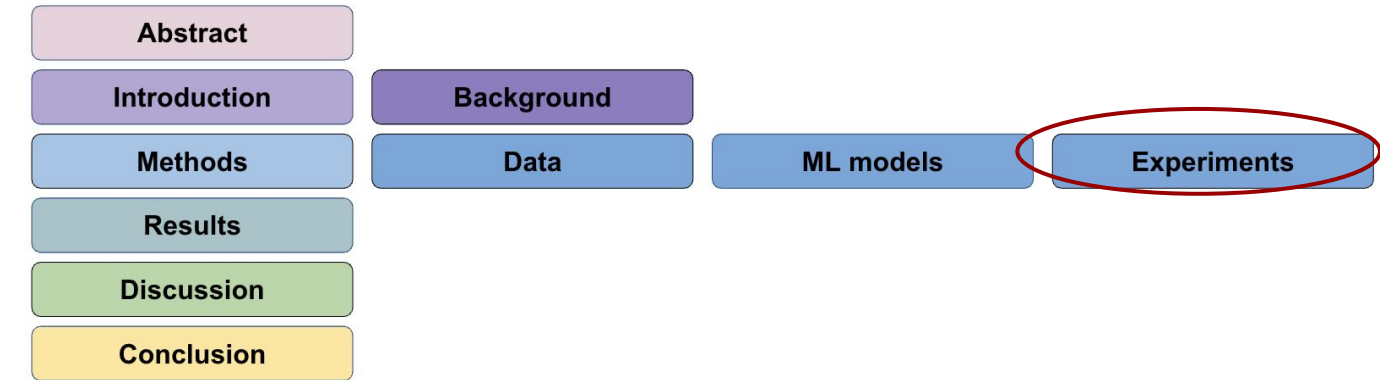
Precision

Recall

Accuracy

F1-Score

AUC-ROC curve



Average Precision (AP) – Primary Metric

- Area under the Precision-Recall curve per class
- Used because of significant class imbalance (Category a = 0.6%) where accuracy alone is misleading
- Other metrics including accuracy, f1 score, kappa, PR and ROC curve, and confusion metrics are used.
- Comparison of DL methods with each other and DL methods with radiologists are both included

Evaluation methods

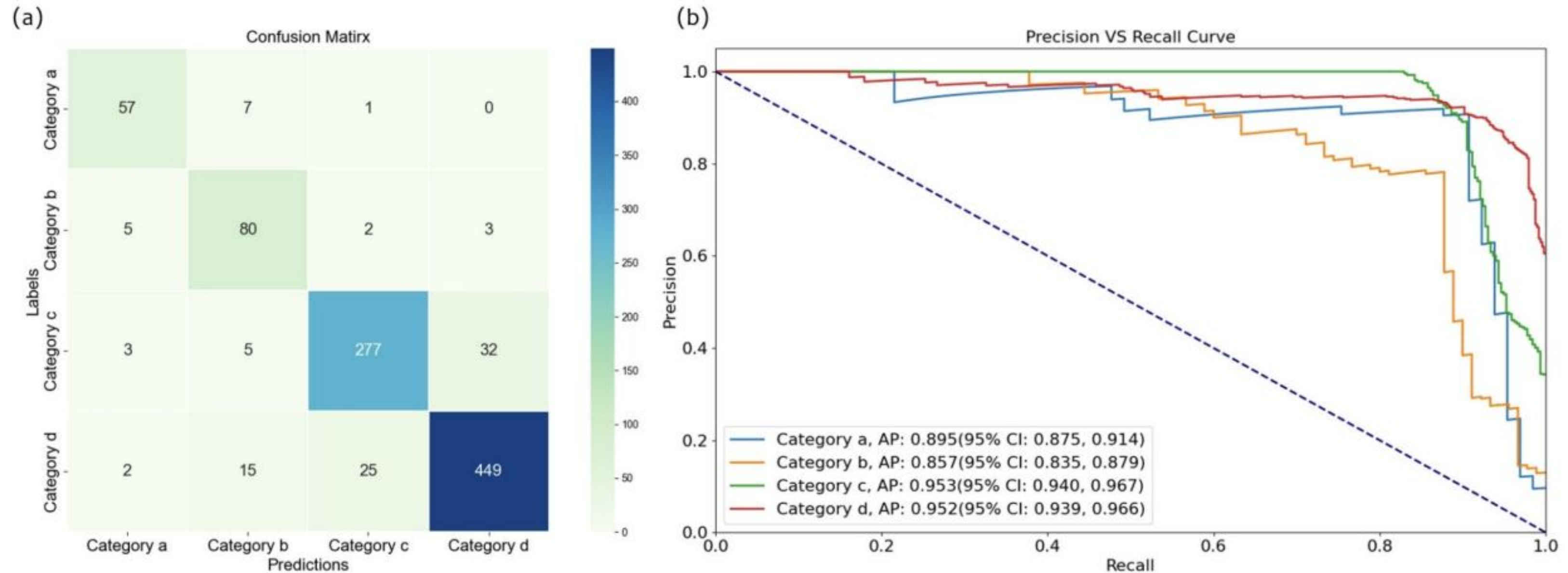


Fig. 2. (a) Confusion matrix of InceptionV3 model on the Taizhou dataset. (b) The Precision-Recall curve of InceptionV3 model on the Taizhou dataset.

Evaluation methods

	Data set	Category a	Category b	Category c	Category d
VGG19	Training	0.999	0.998	0.999	0.998
	Validation	0.890	0.857	0.851	0.902
	Test	0.883	0.828	0.824	0.922
InceptionV3	Training	1.000	0.999	0.999	0.998
	Validation	0.921	0.904	0.894	0.875
	Test	0.987	0.965	0.931	0.911
ResNet50	Training	0.999	0.999	0.999	0.999
	Validation	0.857	0.849	0.901	0.858
	Test	0.869	0.801	0.952	0.846
Xception	Training	0.999	0.998	0.999	0.998
	Validation	0.857	0.872	0.882	0.878
	Test	0.712	0.875	0.889	0.874

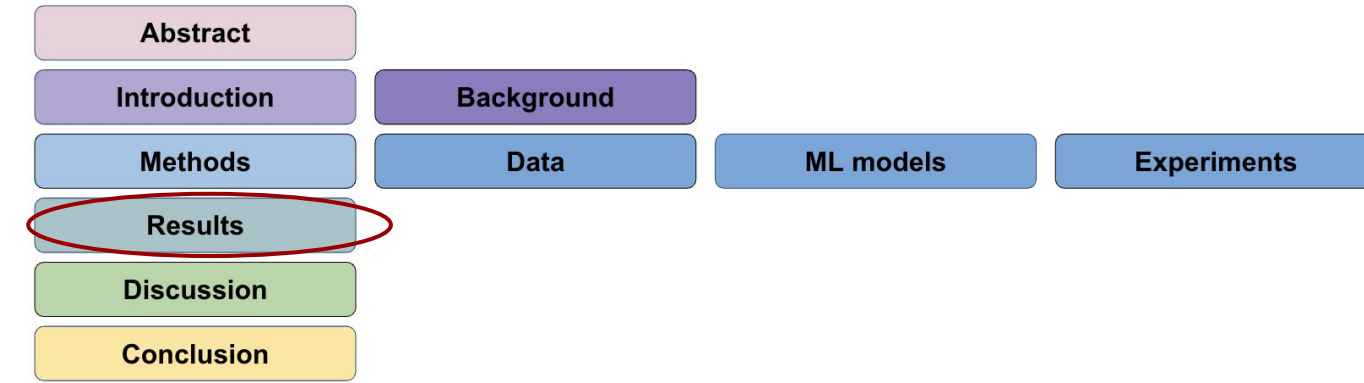
Evaluation methods

	Time	Type	Accuracy (95% CI)	Precision (95% CI)	Recall (95% CI)	F1-Score (95% CI)	Kappa (95% CI)
InceptionV3	0.027s	Category a	0.992 (0.984, 1.000)	0.967 (0.935, 0.998)	0.906 (0.815, 0.997)	0.935 (0.914, 0.957)	0.831 (0.798, 0.863)
		Category b	0.930 (0.908, 0.952)	0.538 (0.417, 0.660)	0.875 (0.773, 0.977)	0.667 (0.625, 0.708)	
		Category c	0.960 (0.943, 0.977)	0.981 (0.965, 0.997)	0.902 (0.858, 0.946)	0.940 (0.919, 0.961)	
		Category d	0.906 (0.880, 0.932)	0.922 (0.889, 0.956)	0.890 (0.851, 0.928)	0.906 (0.880, 0.931)	
Radiologist 1	13.5s	Category a	0.980 (0.968, 0.992)	0.286 (0.049, 0.522)	1.000 (1.000, 1.000)	0.444 (0.400, 0.488)	0.071 (0.048, 0.094)
		Category b	0.836 (0.804, 0.869)	0.288 (0.173, 0.404)	0.298 (0.179, 0.417)	0.293 (0.253, 0.333)	
		Category c	0.491 (0.447, 0.535)	0.437 (0.375, 0.499)	0.482 (0.417, 0.548)	0.459 (0.415, 0.503)	
		Category d	0.559 (0.515, 0.602)	0.486 (0.413, 0.559)	0.407 (0.342, 0.473)	0.443 (0.399, 0.487)	
Radiologist 2	18s	Category a	0.992 (0.984, 1.000)	0.000 (0.000, 0.000)	0.000 (0.000, 0.000)	0.0 (0.000, 0.000)	0.06 (0.039, 0.081)
		Category b	0.872 (0.843, 0.902)	0.434 (0.301, 0.567)	0.404 (0.276, 0.531)	0.418 (0.375, 0.461)	
		Category c	0.481 (0.437, 0.525)	0.453 (0.403, 0.503)	0.772 (0.717, 0.827)	0.571 (0.528, 0.614)	
		Category d	0.545 (0.501, 0.589)	0.409 (0.290, 0.528)	0.125 (0.081, 0.169)	0.192 (0.157, 0.227)	
Radiologist 3	14.4s	Category a	0.984 (0.973, 0.995)	0.300 (0.016, 0.584)	0.750 (0.326, 1.174)	0.429 (0.386, 0.472)	0.069 (0.047, 0.091)
		Category b	0.866 (0.836, 0.896)	0.404 (0.270, 0.537)	0.368 (0.243, 0.494)	0.385 (0.342, 0.428)	
		Category c	0.481 (0.437, 0.525)	0.442 (0.387, 0.497)	0.612 (0.548, 0.675)	0.513 (0.469, 0.557)	
		Category d	0.551 (0.507, 0.594)	0.465 (0.379, 0.551)	0.278 (0.218, 0.338)	0.348 (0.306, 0.390)	

Table 3. Performance of InceptionV3 model compared with radiologists in assessing four categories of mammographic breast density (500 mammograms from the test set of the Taizhou dataset).

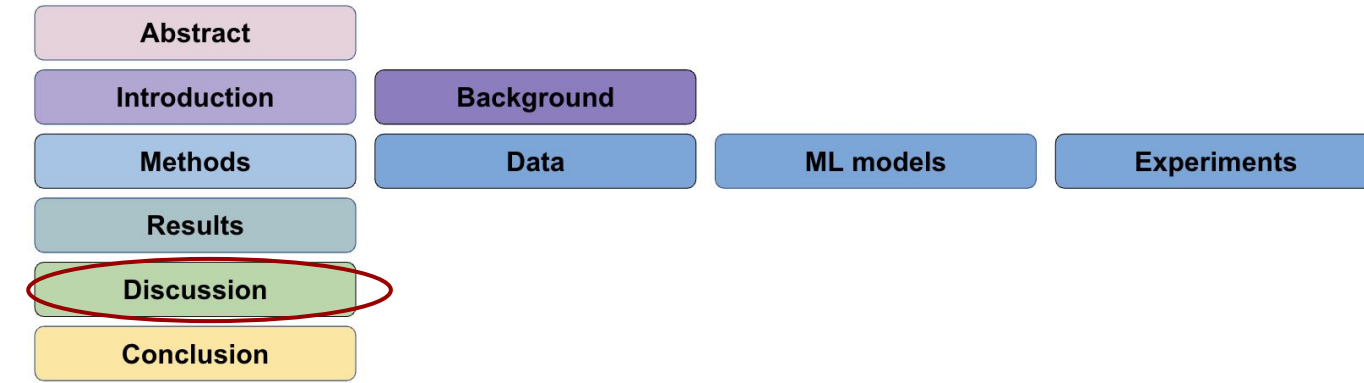
Results

- **Are the results reproducible?**
- **Are the results valid statistically?**
- **Are they explainable?**
- **Did they confirm their hypothesis?**
- **What are their limitations?**

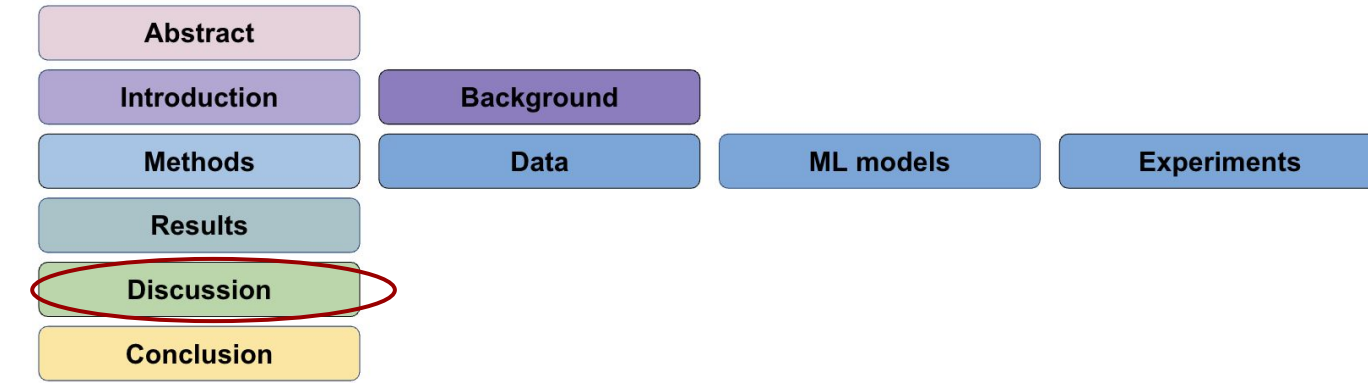


Discussion

- Are limitations acknowledged?
- Is future work discussed?



Details of discussion



Any good researcher should be aware of their study's limitations.

And any good paper should declare them.

Acknowledged limitations:

- Class imbalance
- No clinical context for radiologists
- Age bias

Could also acknowledge:

- No explainability/interpretability
- Single institution data
- unfair radiologist comparison (junior radiologists)
- No discussion of failure cases

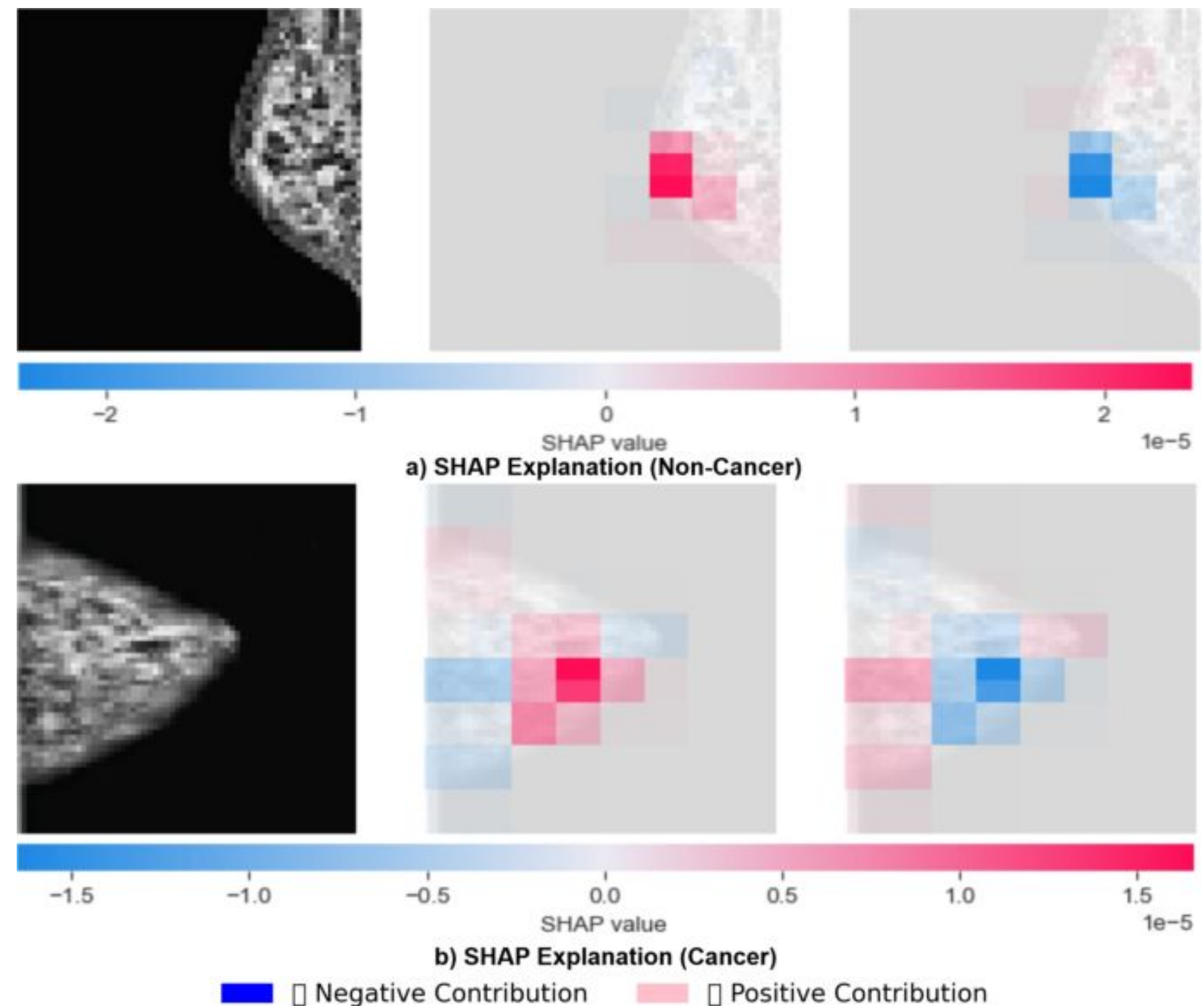
Example of Explainability

Ahmed, J., Ahmed, F., Kabir, M.
A., Ahad, M. T., Jadoon, M. A.,
Rehman, A. ur, & Bermak, A. (2025).

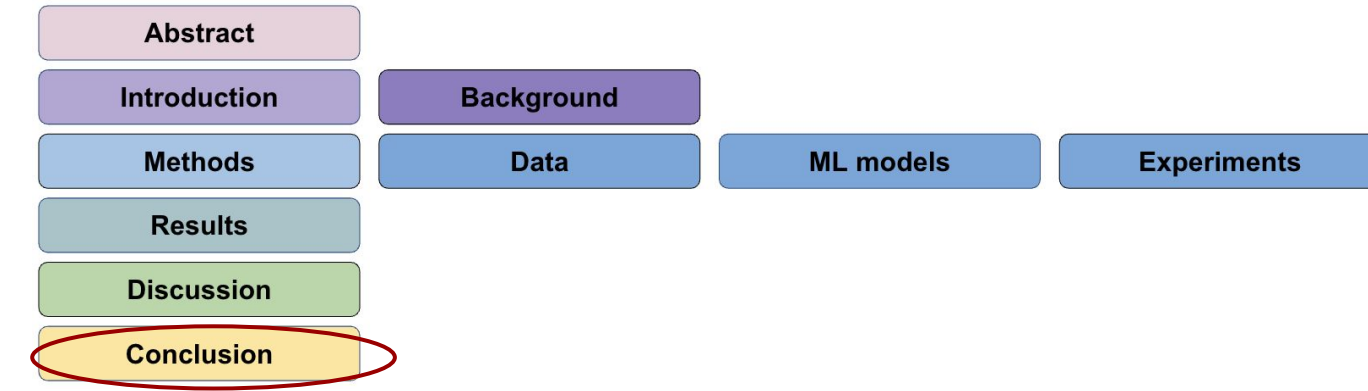
**LBNet: an optimized lightweight
CNN for mammographic breast
cancer classification with
XAI-based interpretability.**

Scientific Reports.

<https://doi.org/10.1038/s41598-025-31642-6>



Conclusion



- Do the claims match the experiments?

Feedback/Attendance



How to build on prior work

- **Search for recent papers using Google Scholar,**
- **Skim their related work sections to find key references**
- **Identify top researchers and their latest papers**
- **Scan top conference proceedings for recent developments**
- **Analyze gaps and limitations:**
 - **Identify unanswered questions, weaknesses, or areas for improvement in existing research**

Tools and resources for reading a paper

- **Relevant Journals and Preprints**

- Arxiv, Bio Arxiv
- BMJ and Nature journals

- **Open source code**

- Github
 - many labs make their projects open source on github
- Papers with code ([Papers With Code](#))

AI tools for reading a paper



ChatGPT
ChatBot



NotebookLM
AI notebook

AI tools – considerations

- Can produce unreliable or incorrect answers – always double-check!
- Image understanding is flawed – see this paper: [Vision language models are blind](#)
- Reading charts require extra caution

Think of the AI tool as a companion you can think out loud with - and don't forget to supervise and double check!



Reminder:

Assignment Deadline: April 5th

Reach out with your questions!



Thank you!